
Musical Agent Systems: MACAT and MACataRT

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Abstract

Our research explores the development of musical agents—artificial systems that automate creative tasks such as composition and improvisation using AI and multi-agent systems. These agents can learn, adapt, and collaborate with human musicians in real-time, employing techniques like corpus-based concatenative synthesis to maintain expressive nuances in performance. We introduce MACAT and MACataRT, two systems designed to enable interactive, human-in-the-loop generative AI to expand creative possibilities for musicians in co-creative settings.

1 Background and Motivation

A **musical agent** [1] is an artificial system that automates musical creative tasks, such as composition and improvisation, using principles from Artificial Intelligence (AI [2]) and Multi-Agent Systems (MAS [3]). These agents operate autonomously, making decisions and performing actions based on real-time musical contexts. These systems range from simple rule-based models to more advanced frameworks capable of learning and adapting through interactions. They are often designed using MAX/MSP programming [4] to facilitate collaboration with human musicians or other agents, exhibiting capabilities such as reactivity, adaptability, and coordination. Another technique employed alongside musical agents is Corpus-Based Concatenative Synthesis (CBCS), which was developed in IRCAM’s CataRT system [5] and has been widely utilized in the field of electroacoustic music [6,7,8]. CBCS utilizes small, personalized datasets of recordings to generate new content by selecting and transforming audio segments from a pre-existing corpus, allowing for real-time control over the generative process and maintaining the expressive nuances of human performance.

Our research is motivated by the aim of developing interactive, human-in-the-loop generative AI systems designed for musicians and sound artists. The primary objective is to enable musicians to explore novel creative possibilities and expand their musical and artistic practices, particularly within real-time scenarios and interactive environments. Our musical agent systems, MACAT and

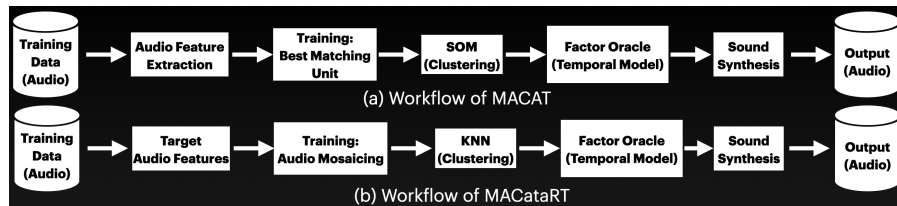


Figure 1: Workflow of our musical agent systems: (a) MACAT and (b) MACataRT.

MACataRT, are based on the mindset of using small data in music [9] and the principles of model crafting in visual arts with generative AI [10], as well as fostering music-making in co-creative spaces between human musicians and musical agents [11], including machine listening [12].

2 Musical Agent System

MACAT, as workflow displayed in Figure 1(a), is an enhanced version of the Metacreation Lab’s musical agent, MASOM [13]. MACAT is an advanced version of MASOM featuring integrated real-time sound synthesis and visualization. It employs concatenative sound synthesis as its primary functionality and includes improved visualization for past and current nodes. Each audio segment generated by MACAT can be easily synthesized according to the musician’s specific needs, allowing sound artists to explore an extensive range of timbral possibilities for each sound sample using sound design parameters. MASOM utilizes Self-Organizing Maps (SOM [14]) and a temporal model called the Factor Oracle (FO [15]), a suffix automaton used for real-time pattern recognition within sequences of nodes. These nodes represent clusters of audio segments grouped by timbral similarity based on extracted audio features. During MACAT’s training, the system undergoes an offline machine listening process where the original audio (training data) is analyzed, and the sequence of nodes is learned. The Best Matching Unit (BMU) plays a key role in SOM training, identifying the node that most closely aligns with an input vector using Euclidean distance as a similarity measure. The SOM training involves initializing nodes, finding the BMU for each input, and adjusting the BMU and its neighbours to match the input vector better. Once trained, the SOM clusters data into groups represented by its nodes. MACAT then adapts its generative output according to its self-listening process and a congruence parameter, which controls the likelihood of FO’s forward and backward transitions to determine the system’s next state.

MACataRT, as workflow shown in Figure 1(b), is an enhanced version of IRCAM’s CataRT system [1], integrating a temporal model based on the factor oracle and offering a more intuitive interface for sound synthesis and resampling. The original CataRT system is limited to reactive improvisation based on real-time machine listening. It clusters audio segments using K-Nearest Neighbors (KNN [16]) and targets specific audio features to perform audio mosaicing, allowing it to listen and react accordingly. Consequently, this system lacks a temporal model for handling time-based musical structures. To address such limitation, we integrate a temporal model by incorporating the factor oracle to automate the generation process. MACataRT is developed on the foundation of CataRT’s audio mosaicing capability. Audio mosaicing [17] is a technique that involves assembling audio segments to form a new audio piece, selecting segments based on their similarity to the target audio features as selected by musicians, who listen to the output to replicate the characteristics of the target audio. MACataRT contains interactive audio mosaicing that operates in real-time and offline. It enables musicians to play along with MACataRT, and it will reactively improvise based on real-time machine listening with target audio features without FO (reactive improvisation). MACataRT also operates in a mode where it learns the sequences of indices of audio segments played during the offline training process, allowing the FO to improvise based on these learned sequences of audio segments (proactive improvisation). Additionally, Appendix A offers a detailed explanation of the latest version of MASOM by us, and Appendix B explains the interface of MACAT and MACataRT.

3 Musical Application

Each musical agent’s application varies, as each provides different musical affordances. MACAT is particularly suited for solo performances, adapting its improvisational output through a self-listening process that allows the agent to lead in most musical settings. In contrast, MACataRT offers greater versatility through audio mosaicing, enabling a wider range of creative possibilities in reactive and proactive improvisation. The effectiveness of MACataRT in practical musical scenarios was proven with a human percussionist (Keon) and guitarist (Sara), as it achieved recognition as one of the Top 10 finalists in the 2024 AI Music Song Contest [18] with the song *Echoes of Synthetic Forest (KeRa)*.

4 Conclusion

In conclusion, our research demonstrates the potential of musical agents to enhance creative collaboration between AI and human musicians. By leveraging advanced techniques such as corpus-based concatenative synthesis, these agents can learn and adapt in real-time, preserving expressive nuances in performance. The introduction of MACAT and MACataRT highlights how interactive, human-in-the-loop generative AI can broaden musicians’ creative horizons, fostering innovative co-creative environments. Appendix C addresses the ethical considerations of generative music AI systems.

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